Mapping Cardinality

E/R Diagrams & Defining table relationships

Chen notation reminder - Before we begin…

Element	Symbol	Meaning
Entity	▭ Rectangle	Represents an entity type
Attribute	◯ Oval	Describes a property of an entity
Key attribute	Underlined within oval	Uniquely identifies each entity instance
Relationship	◇ Diamond	Represents an association between entities
Arrows / lines	→ Connecting lines	Indicates an attribute OR a relationship
Inheritance	▽ Inverted triangle	Represents a parent-child relationship

Mapping Cardinality

Mapping cardinality in an E/R model describes how many instances of one entity can be associated with instances of another entity.



Cardinality = Relationships

Cardinality indicates the relation between entities (tables) and their records/tuples (rows).

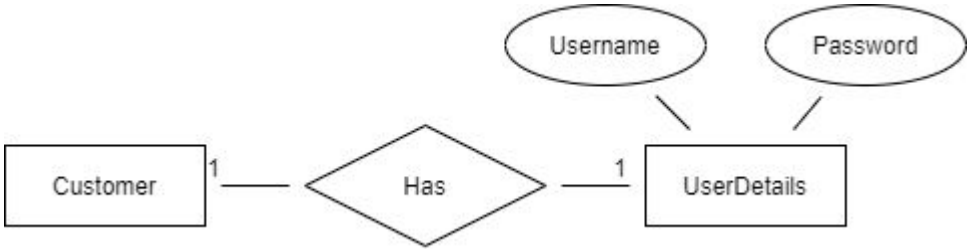
Two given tables will result in up to three types of connectivity:

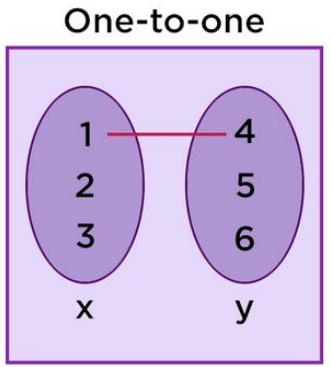
- One to one (1:1)
- One to many (1:N) or Many to one (N:1)
- Many to many (N:M)



Mapping Cardinality - One to One (1:1)

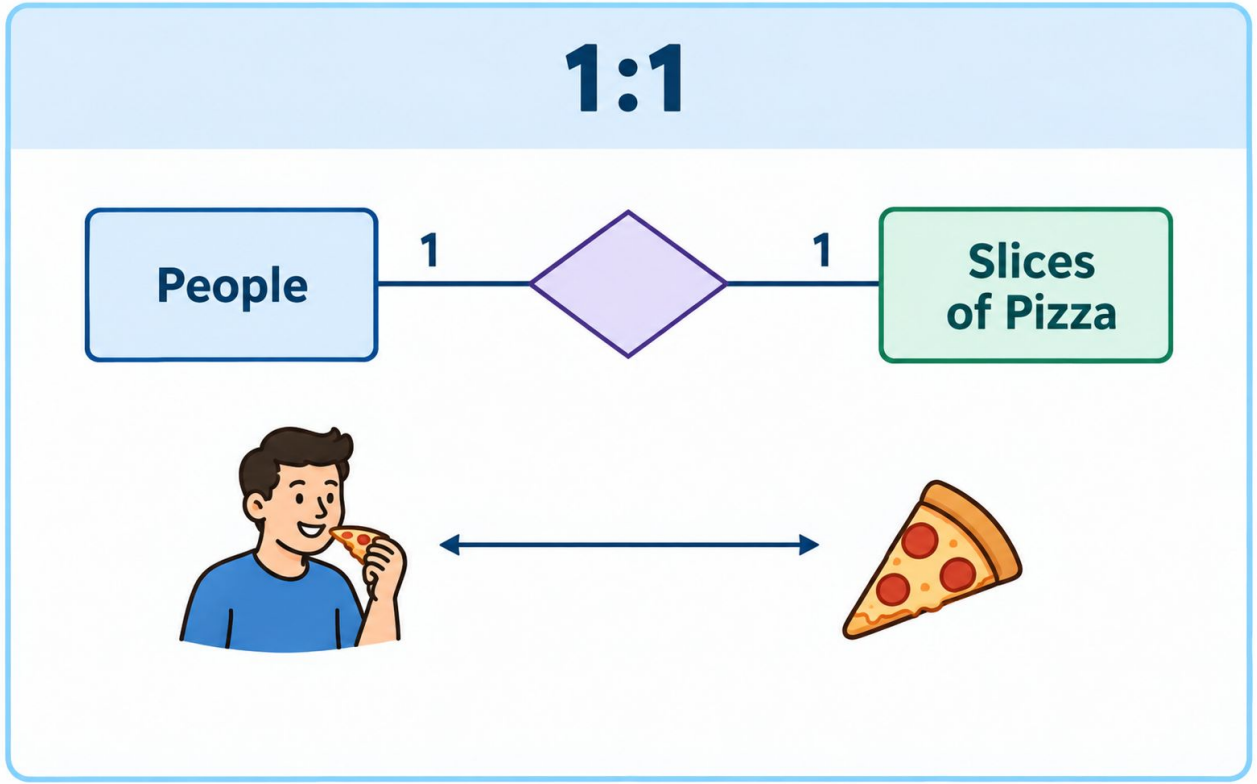
One-to-one means that for the two entities connected by the line, **there is exactly one tuple of the first entity for every one tuple from the second entity**.

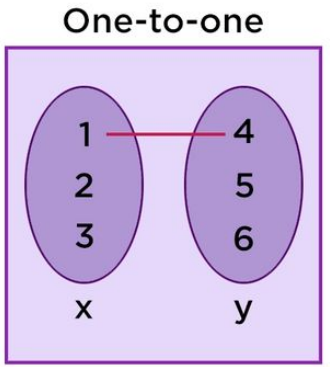




Pizza party - One to One (1:1)

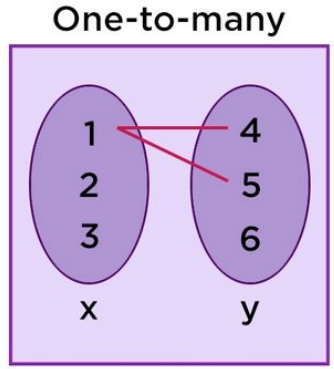
Only one slice of pizza per person!





Mapping Cardinality - One-to-many (1 : N)

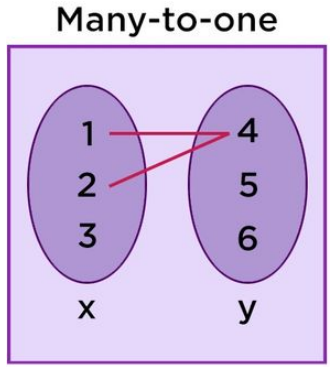
A **tuple in the first** entity can be associated with **any number** of tuples **from the second** entity, BUT a **tuple from the second** entity can **only** be associated with **one tuple from the firs**t entity.





Mapping Cardinality - Many-to-one (N : 1)

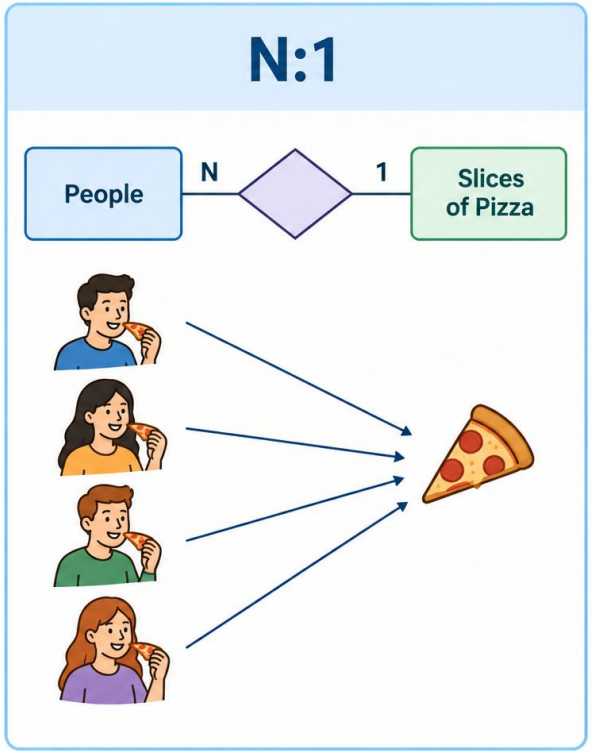
A **tuple in the first** entity can **only** be associated with **one tuple from the second** entity, BUT a **tuple from the second** entity can be associated with **any number** of tuples **from the first** entity.

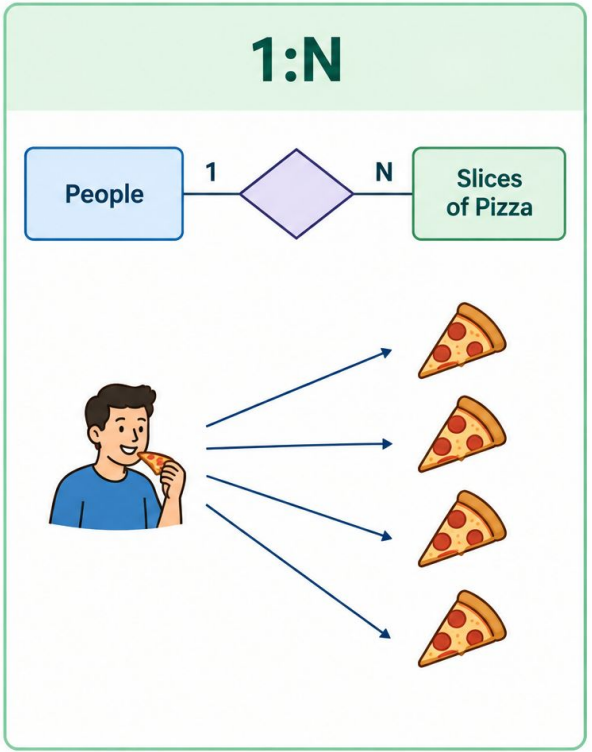




Pizza party - Many to one (N:1) and One to many (1:N)

Only one slice of pizza per person... BUT they can be SHARED!



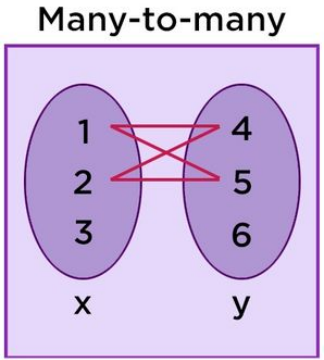


People can eat as many slices as they want AND will NOT share them!

Mapping Cardinality - Many-to-many (N : M)

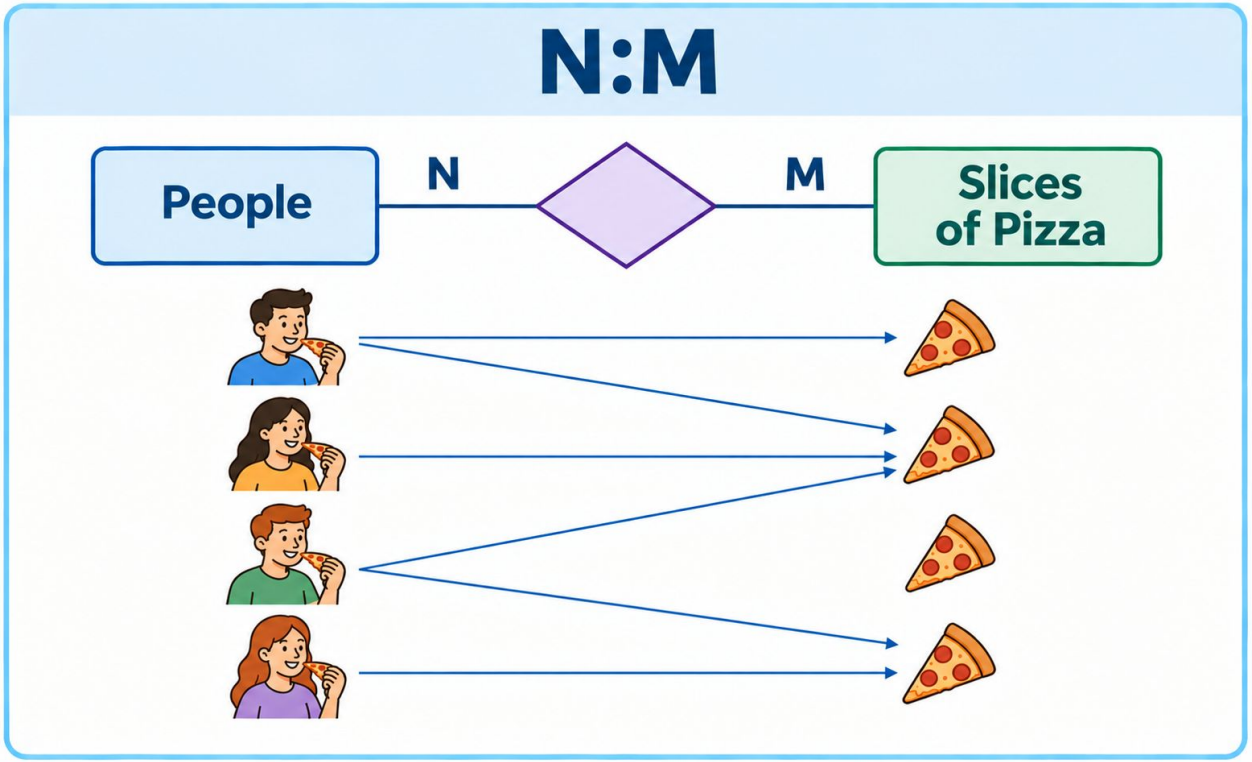
For two entities connected, **each tuple of the first** entity can be related to **one or more** tuples **from the second** entity and, from the other perspective, **each tuple of the second** entity can be related to **one or more** tuples **from the first** entity.

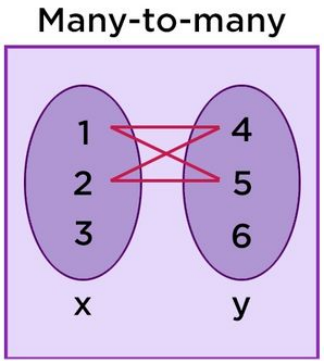




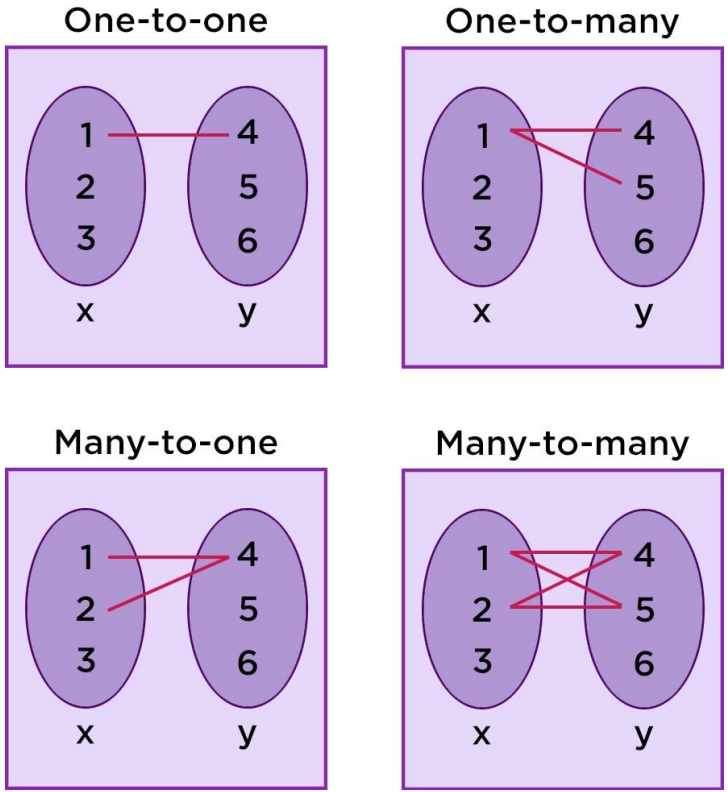
Pizza party - Many to many (N:M)

Everyone can have as many pizza slices as they want AND share them!





Mapping Cardinality - When in doubt…!



Mapping Cardinality - Min-Max Notation

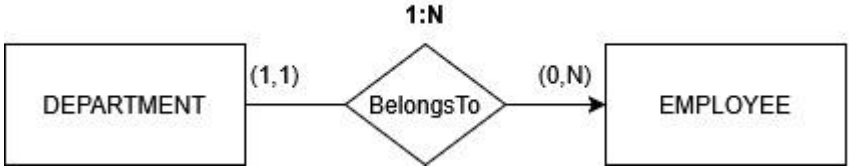
Min-Max Notation (or Minimum-Maximum Cardinality Notation) specifies the **minimum and maximum number of tuples of one entity** that can participate in a relationship with another entity.

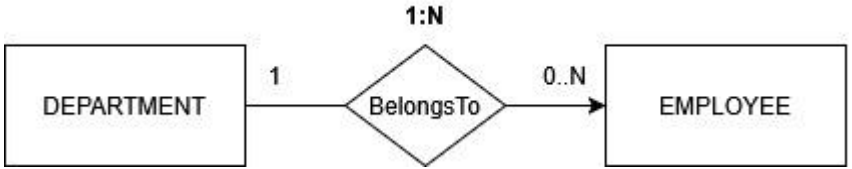
- **(1,1)** or 1 → exactly one (there will always be one, and only one)
- **(0,1)** or 0..1 → zero or one (there can be none or one)
- **(0,N)** or 0..N → zero or many (there can be none, one or many)
- **(1,N)** or 1..N → one or many (there will always be one or more)

The 0..1 format is more commonly associated with UML notation.

Mapping Cardinality - Min-Max Notation (Example)

So, 1:N describes the overall mapping cardinality, while **(min,max) describes the participation constraints** more precisely.

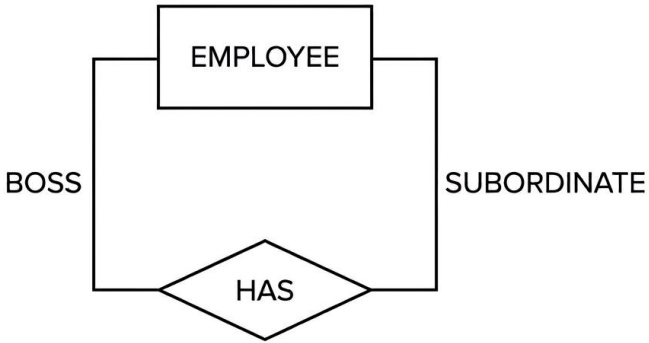




Other relationships - Recursive (aka Reflexive)

A recursive interrelationship associates **tuples from one entity with other tuples of the same entity**.

Only one entity is involved.





Other relationships - Recursive (aka Reflexive)

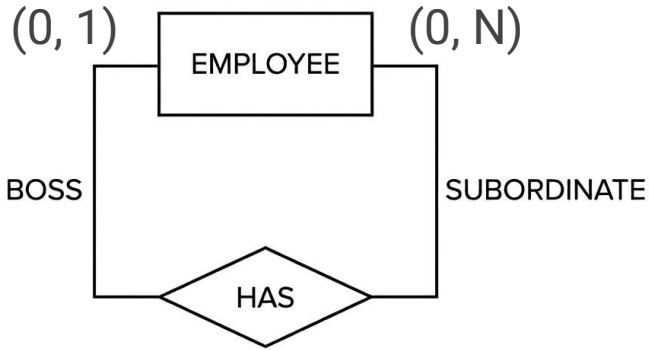
The Employee entity set participates in **two role**. In this example: Supervisor (Boss) and Subordinate. These are called role names.

For the Boss role:

- The **minimum** cardinality is **0** because the lowest-level employee (a subordinate) may not manage anyone.
- The **maximum** cardinality is **N**, as an employee can manage many subordinates.

For the Subordinate role:

- The **minimum** cardinality is **0**, as the CEO, for example, is not a subordinate to anyone.
- The **maximum** cardinality is **1**, as a subordinate can have only one manager.

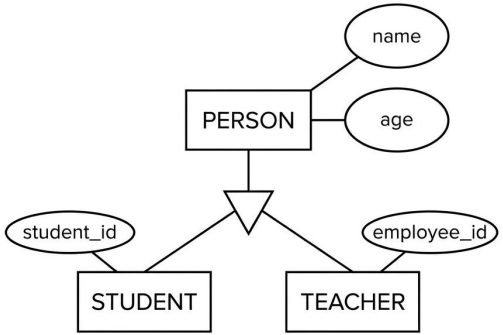


Other relationships - Inheritance

Inheritance allows a more **specific entity** to **share** the **common attributes of** a more **general entity**.

This **avoids repeating shared attributes** while allowing each derived entity to have its own specific attributes.

It can be understood as a father-child relationship.



Mapping Cardinality - Diagram Example

A company wants to design a database to manage its **departments** and **employees**.

- Each department has a unique identifier (**DeptID**), a **name**, and a **room**.
- Each employee has a unique identifier (**EmployeeID**), a **name**, a **surname**, a **salary**, and a **hire date**.
- Each employee **belongs to exactly one** department, while a department can have **zero or more** employees.
- The company also wants to record the boss of each employee. An employee can have **zero or one** boss, while a boss can supervise **zero or more** employees.
    - There will only be one case where an employee has no boss, can you guess who is he?

Mapping Cardinality - Diagram Example

